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Integration of oscillator-based feature extraction for energy-efficient convolutional neural networks

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ABSTRACT

The rapid growth in machine learning and artificial intelligence workloads has increased the demand for computing power. Although digital computing accelerators dominate today's market, analog computing architectures are emerging as promising energy-efficient alternatives. In this work, we propose an approach to address these challenges by replacing the first layer of a convolutional neural network (CNN) with a network of oscillatory retinal neurons (ORNs) composed of weakly coupled, optically activated negative differential resistance (NDR) devices. Each ORN consists of a photodetector exhibiting NDR behavior under illumination, coupled with an inductor forming a self-oscillating circuit without needing external voltage sources. We model the nonlinear oscillator dynamics using experimentally measured device characteristics and simulate their behavior under varying optical inputs. Simulations performed on the Fashion-MNIST (Modified National Institute of Standards and Technology) dataset demonstrate that the hybrid ORN–CNN architecture maintains high recognition accuracy and achieves 95% training accuracy and 92% testing accuracy comparable to the fully software-defined CNN with 93% test accuracy while dramatically reducing energy consumption to 24 aJ/operation. Different ORN network topologies were investigated, with asymmetric inductive coupling achieving consistently high performance. These results highlight the potential of oscillator-based networks for low-power analog computing and demonstrate the possibility of using nonlinear physical systems for energy-efficient machine learning applications. This work demonstrates a possible path for integrating device-level oscillator dynamics into future neuromorphic and analog computing platforms.

INTRODUCTION

CNNs have been widely used as an architecture for image-processing tasks such as image recognition.[1–3] CNNs resemble artificial neural networks (ANNs) as they are composed of neurons that self-optimize through learning. Although CNNs are renowned for their precision, they are limited by computational resources, particularly in terms of speed and power consumption.[4–6] CNNs operating on traditional von Neumann architectures face these challenges primarily due to the extensive memory access required to train and update numerous parameters for each layer of a deep network.[4,7,8]

The emergence of hardware inspired by neural structures provides the potential to accelerate these algorithms by utilizing in-memory computing paradigms, subsequently reducing memory consumption.[9–13] Neuromorphic computing systems employ a variety of approaches and technologies, such as spiking neural networks (SNNs) and oscillatory neural networks (ONNs).[14–19] ONNs, structured as a network of coupled oscillators connected by synaptic weights, try to mimic the collective synchronized behavior observed in human brains. ONNs represent an analog-based computing paradigm that encodes information through oscillation phase or frequency.[20–26] Implementing parallel oscillators' processing information in frequency and phase domains leads to fast and low-power computations.[27–29] The relationship between CNNs and the associative memory capabilities of ONNs in image-processing applications has been widely explored and they have been used as hardware accelerators in CNNs.[30–37] Time-encoded output signals of four-coupled vanadium oxide oscillators can store up to five trained filters performing the equivalent function of convolutional filters. Hence, five convolutional filters can be trained on a single ONN platform with an energy consumption of 3.4 μJ per frame. The integrated network exhibits a test accuracy of 95% on the MNIST (Modified National Institute of Standards and Technology) dataset while the original convolutional neural network (CNN) shows 97% accuracy. The worsening of test accuracy is attributed to the failure of the neural network to provide feature edge extraction.[4]

We investigate the capabilities of a network of coupled oscillators to provide a possible alternative to CNNs. Our research shows the potential of replacing the first convolutional layer of a CNN with a network of 3×3 ORNs. Our network of ORNs is composed of nine photosensors converting incident light into voltage signals capable of performing simultaneous calculations on the input data. Each sensor is an oscillator representing different voltage signals and exhibiting different frequency spectra based on the optical power received. We show how the generated frequency spectrum through a network of coupled oscillators can perform image-processing tasks like feature extraction. Each photosensor of the 3×3 oscillator array acts as one of the pixels of a 3×3 image pattern. The feature extraction capability of the ORNs was carried out through the simulation of oscillator dynamics. The oscillators, which compute in parallel at different frequency bands, provide an improved speed compared to traditional von Neumann architectures. Multiple steps have been taken to replace the first convolutional layer of a CNN with a network of coupled oscillators. The CNN is structured with two convolutional layers, each with 32 and 64 filters, respectively. Initially, 32 feature maps of the first convolutional layer (Conv1) of a trained CNN have been extracted at the final epoch of simulation, where the accuracy saturates to its maximum value. Furthermore, these extracted feature maps from CNN will be compared with the generated feature maps by a 3×3 network of ORNs. We have used a structural similarity index matrix (SSIM) to compare the feature maps of oscillatory retinal neuron (ORN), and 32 feature maps created by Conv1 3×3 filters. A comprehensive explanation of SSIM and our comparison methodology to identify the feature maps most similar to the traditional Conv1 outputs is discussed in Sec. S1 in the supplementary material. After showing that our ORNs can extract the feature maps and replace the output of the first convolutional layer, we will proceed to investigate the effect of substituting an energy-efficient oscillatory network with a layer of a fully software-defined CNN. The evaluation metric we adopt for the performance analysis of the combined ORN–CNN structure is the recognition accuracy. Exhibiting a very low energy consumption related to peripheral circuits and providing parallel computation, these oscillators have the potential to address the limitations of replacing the convolutional layers of a CNN. We systematically evaluate 30 oscillator topologies, identify asymmetric inductive coupling as the optimal architecture, and demonstrate that a single ORN topology can replace all 32 Conv1 filters while maintaining comparable accuracy on the Fashion-MNIST (Modified National Institute of Standards and Technology) dataset. We also present a multi-scenario energy analysis providing a roadmap for progressive hardware integration.

METHODOLOGY

The ORN network comprises weakly coupled oscillators that can be used as computing units, similar to kernel filters, to perform convolution operations that are computationally expensive. We demonstrate a novel approach that substitutes the linear mathematical transformations used in CNNs to generate feature maps with ORNs that implement nonlinear transformation on the input image. A linear convolutional operation involves sliding a kernel filter over an input image, computing the dot product of the filter with each pixel of the input image it covers at different positions. Different filters extract diverse features, such as shapes and edges, from the input image. The result of the convolution operation applied to an input image is a feature map that demonstrates the detected features by the filters. The convolution operation provides a way to convert the input data into a form that exhibits the essential features of input data for further processing and applications. With convolution operations being a linear transformation operation, the spatial relationship within the data will be preserved.

An oscillatory network performing nonlinear transformation can also generate feature maps that exhibit specific characteristics of the input data, enabling an image-processing application. The oscillatory network is designed to be an array of optically activated negative differential resistance (NDR) devices operating as oscillators. The interaction between weakly coupled NDR neurons modulates the frequency spectrum band as a nonlinear outcome of the input light intensities exposed on various neurons. ORNs generating multiple feature maps of an input image at distinct oscillation frequencies and amplitudes lead to encoding image features within the neuron's frequency spectrum. Figure 1(a) shows the circuit schematic of an oscillatory neuron. The ORN unit comprises a photodetector that exhibits voltage-controlled NDR behavior and an inductor that forces the device to oscillate when exposed to light. The inductive element can be implemented using a Hara active inductor, which consists of a single MOSFET and a resistor, providing a compact on-chip alternative to passive coil-based inductors.[38] The device is composed of a semiconductor-graphene-metal (SGM) photodetector with NDR characteristics during light illumination as described in detail in Ref. 38 along with its fabrication process, device physics, and ultra-low power characteristics. In this work, we build on these device capabilities and investigate the potential of ORNs for replacing the power-intensive convolutional layers of CNNs. Figure 1(b) shows the experimentally measured J–V characteristics of a $1 \times 1\ \mathrm{mm}^2$ optically activated NDR device under an optical power density of $3\ \mathrm{mW/mm}^2$.[38] As it is shown, the device does not require an external voltage source to establish an NDR behavior since the negative slope of the I–V curve from negative voltage values. According to the circuit schematic of the ORN, the device has an intrinsic capacitance measured experimentally to be around 3 nF.[38] The capacitance value was extracted from C–V measurements performed on fabricated SGM photodetectors using a Keysight B1500a semiconductor parameter analyzer under illumination at 445 nm. The capacitance per unit area was determined from measurements on larger devices and scaled to the $1 \times 1\ \mathrm{mm}^2$ device used in our simulations.[38] This capacitance is governed by the parallel-plate geometry of the SGM junction and is expected to scale approximately linearly with device area. The combination of a capacitor and an inductor create an LC tank circuit, which, when forced by the NDR behavior of the optically activated neuron, mimics the behavior of a relaxation oscillator. The emergence of oscillation due to the device's inherent characteristics and the electrical properties of the LC components is analogous to an oscillatory behavior observed in a Van der Pol oscillator.

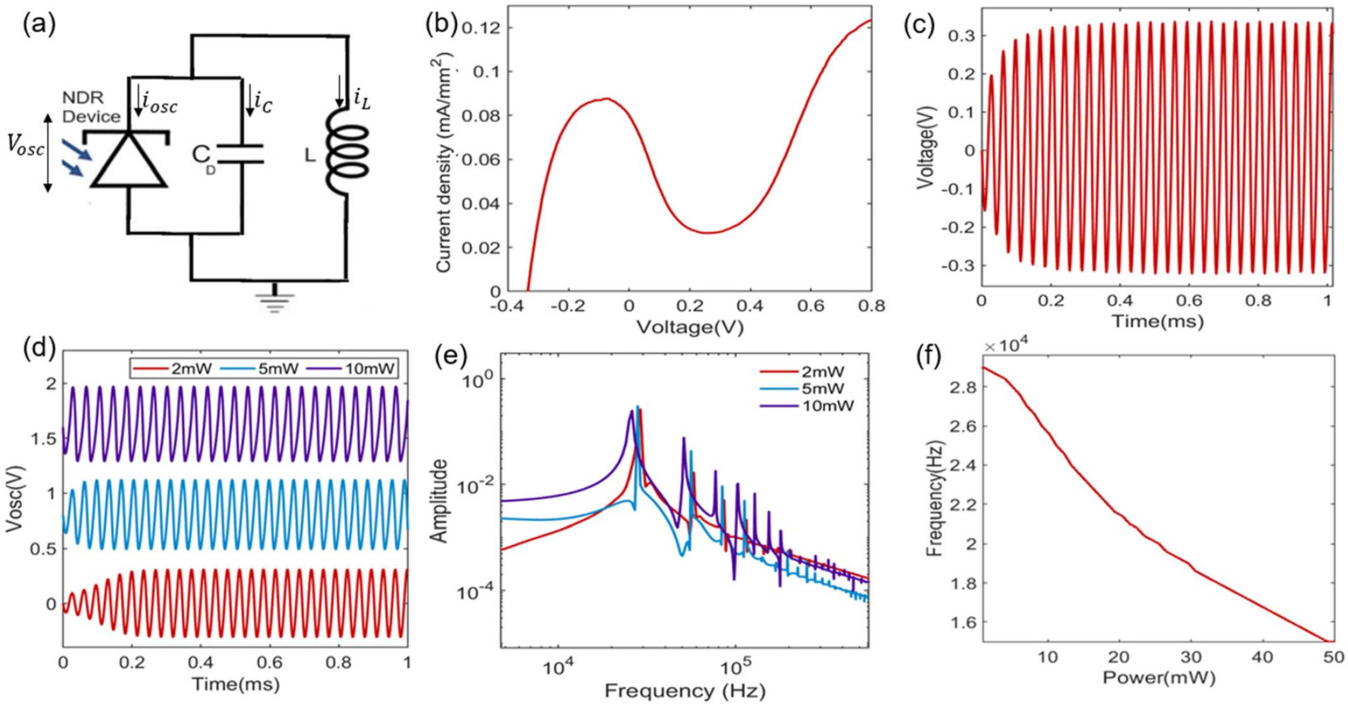

FIG. 1. Oscillatory behavior of an ORN. (a) Schematic of an optically activated NDR device connected to an inductor. (b) J–V curve of one of the NDR devices measured under uniform illumination. (c) Voltage–time oscillation of one oscillator. (d) Simulated oscillation behavior of a single device under different light power illumination. (e) Fourier transform of the V–t curves. (f) Frequency of oscillation as a function of light power.

Van der pol oscillators follow the below equation, where x represents the position variable, t is the time, and μ is the damping factor. When μ is zero, the oscillator acts like a simple harmonic oscillator and with increasing the damping factor, the system exhibits more nonlinear behavior,[39,40]

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0. \tag{1}$$

Similar to a Van der Pol oscillators, our network of oscillators also follow a differential equation,

$$i_C = C\frac{dV_C}{dt}, \tag{2}$$

$$V_L = L\frac{di_L}{dt}, \tag{3}$$

$$i_{osc} = G(V_{osc}, P)V_{osc}, \tag{4}$$

$$V_L = V_c = V_{osc}, \tag{5}$$

$$i_C + i_L + i_{osc} = 0, \tag{6}$$

$$C\frac{dV_C}{dt} + \frac{1}{L}\int V_{osc}dt + G(V_{osc}, P)V_{osc} = 0, \tag{7}$$

$$C\frac{d^2V_{osc}}{dt^2} + G(V_{osc}, P)\frac{dV_{osc}}{dt} + \frac{V_{osc}}{L} = 0. \tag{8}$$

Here, $G(V_{osc}, P)$ is the conductance of the NDR device, which is a nonlinear function of oscillating voltage V_{osc} and incident light power P. The experimentally measured J–V curves serve as the input to the nonlinear conductance function $G(V_{osc}, P)$ in our simulations. C is the intrinsic capacitance of the NDR device that is experimentally measured and is a known quantity same as $G(V_{osc}, P)$. L is an external inductor connected to the device to force oscillations to happen. i_C, i_L, and i_{osc} are the currents through the capacitor, inductor, and the NDR device. V_L and V_c are the corresponding voltages of inductor and capacitor. All subsequent results are derived from circuit-level simulations using these experimentally extracted parameters. Figure 1(c) shows the oscillating behavior of an NDR device connected to an active inductor. Since the device only indicates oscillation under illumination, Fig. 1(c) highlights the fact that voltage–time oscillations are optical intensity dependent and vary with optical power changes. As mentioned earlier, the negative slope of the NDR device J–V curve starts from −100 to 200 mV.[38] This regime of photovoltaic operation defines the required optical power range to derive oscillations. The minimum required light power that allows device to show NDR behavior is arbitrary and does not affect the value of optical conductance. Figure 1(d) shows the corresponding frequency spectrum of oscillations at specific optical powers. Overall, as shown in Fig. 1(e), the frequency of the oscillations decreases with increasing optical power. We will provide an analysis demonstrating the computational capabilities of a network of coupled oscillators. Different network topologies consisting of two coupled oscillators are investigated under light illumination. We need to gain insight into the dynamics of the coupled ORNs and find out how different coupling impedances can change the computational capabilities of the network.

In the model, the experimentally determined I–V curve of the photodetector is implemented to simulate its behavior when connected to an active inductor. Then, the effect of coupling two ORNs under different light illuminations is studied. In this simulation, the pixel intensity range of 0 and 1 is mapped to 2500 different combinations of P_1 and P_2 considering 50 equally spaced points between 0 and 1 for each oscillator and sweeping the light power between 0 and P_{max}. The variation of light exposed to the couped oscillators is reflected on their frequency spectrum by shifts in oscillation peaks and creation of subharmonic peaks. Each V–t oscillation has a corresponding frequency spectrum characterized by different center frequencies and BW. Figure 2(a) depicts the schematic of two inductively coupled oscillators, coupled with a 10 mH inductor. Since both NDR devices are connected to similar value 10 mH inductors to drive the oscillation, the ORNs have a symmetric topology. Employing an optical power range between 0 and P_{max} = 10 mW for both devices and sweeping the power results in the emergence of nonlinear peak surfaces at different center frequencies. Figure 2(b) shows the voltage heatmap of two coupled ORNs showing the magnitude of the bandpass filtered oscillations with changing P_1 and P_2 at 28 200 Hz center frequency for different coupling inductances L_c. Results from the simulation represent circular-shape curves for different values of coupling inductor. These symmetrically curved subspaces created with optical variation of two inductively coupled oscillators with a 200 Hz BW, demonstrating variability across different coupling impedances. The bandpass filter bandwidth of 200 Hz was selected after testing a range of bandwidths from 50 Hz to 1 kHz in our simulations to balance spectral selectivity and signal energy in the extracted feature maps. This bandwidth is consistent with the observed spacing of dominant oscillation peaks in the simulated frequency spectra. Narrower bandwidths (<100 Hz) produced excessively sparse feature maps with insufficient spatial information, while wider bandwidths (>500 Hz) merged adjacent spectral peaks, reducing feature discriminability. Using different values of L_c results

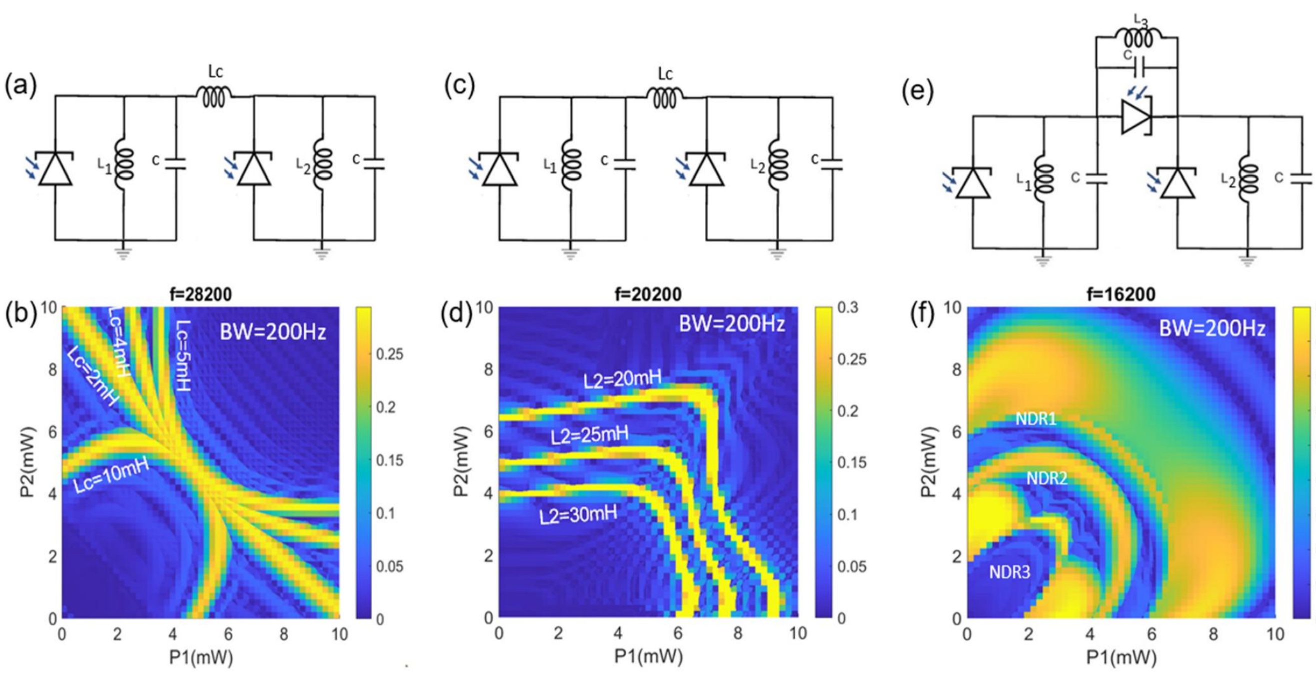

FIG. 2. Computation using the frequency spectrum of the ORNs. (a) Circuit schematic of two inductively coupled ORNs, both NDR devices are connected to similar inductors for oscillation creating a symmetric topology. (b) ORN voltage colormap showing nonlinear and different peak surface shapes for different coupling impedances. (c) Circuit schematic of two inductively coupled ORNs, each NDR devices are connected to a specific inductor value creating an asymmetric topology. (d) ORN voltage colormap showing nonlinear and different peak surface shapes for different L_2 values. (e) Circuit schematic for two coupled ORNs connected with an NDR device, both NDR devices are connected to similar inductors for oscillation creating a symmetric topology. (f) ORN voltage colormap showing nonlinear and different peak surface shapes for different NDR devices.

in a variation of coupling strength, and hence, change in the subspace shapes. The higher the value of L_c, the weaker the coupling strength. Therefore, 2 mH provides a very strong coupling between two oscillators that leads them to lock-in frequency, and the relationship between their powers tends to behave like a line rather than a curve. Considering the differential equations followed by two coupled oscillators, the effect of implementing various coupling impedances on their behavior can be justified,

$$i_{C_1} + i_{L_1} + i_{osc_1} + i_{L_c} = 0, \tag{9}$$

$$i_{C_2} + i_{L_2} + i_{osc_2} - i_{L_c} = 0, \tag{10}$$

$$V_{osc_1} - V_{osc2} = L_c \frac{di_{L_c}}{dt}, \tag{11}$$

$$C\frac{d^2V_{osc_1}}{dt^2} + G(V_{osc_1}, P_1)\frac{dV_{osc_1}}{dt} + \frac{V_{osc_1}}{L_1} = -\frac{di_{L_c}}{dt} = \frac{V_{osc_2} - V_{osc_1}}{L_c}, \tag{12}$$

$$C\frac{d^2V_{osc_2}}{dt^2} + G(V_{osc_2}, P_2)\frac{dV_{osc_2}}{dt} + \frac{V_{osc_2}}{L_2} = \frac{di_{L_c}}{dt} = \frac{V_{osc_1} - V_{osc_2}}{L_c}. \tag{13}$$

i_{L_c} is the current of the coupling impedance. The equations represent the dependence of various components on an oscillatory network and how the variation of one parameter affects the relationship between other parameters. The change of incident light on one oscillator leads to a change in its voltage oscillation that, due to the existence of the coupling impedance in the network, influences the change in current value and also the other oscillator's voltage value. As shown in the equations, changing the value of coupling impedance L_c from 10 to 5 mH when $L_1 = L_2$, effectively doubles the influence of the coupling on the system dynamics. This is because the term $\frac{V_{osc_1} - V_{osc_2}}{L_c}$ explains the coupling effect and reducing L_c increases the term's magnitude leading to an amplification between the oscillators' interactions. To quantify the coupling strength, we define a dimensionless coupling factor $\kappa = \frac{L_{osc}}{L_c}$, where L_{osc} is the individual oscillator's self-inductance and L_c is the coupling inductance. Weak coupling corresponds to $\kappa \le 1$, where the oscillators maintain distinct natural frequencies and the coupled dynamics produce computationally rich subspaces with diverse, non-trivial shapes. Strong coupling ($\kappa \gg 1$) drives the oscillators toward frequency-locking, for example, $L_c = 2$ mH with $L_{osc} = 10$ mH yields $\kappa = 5$, producing the near-linear subspace observed in Fig. 2(b). The weak-to-moderate coupling regime ($\kappa \le 1$) is exploited throughout this work, as it provides the greatest subspace diversity for feature extraction. The change in coupling impedance value adjusts the ratio of capacitance value, the device's conductance, and the device inductor's value accordingly. Therefore, adjusting L_c not only varies the coupling strength but also influences the overall dynamics of the circuit by modifying the relative contribution of other parameters in the circuit. Thereby, the change in the subspace shapes verifies the topology's effect, which is dictated by the coupling strength and the inductance values associated with each device on the oscillators' frequency spectrum. This modification in the subspace shape translates to the computation of distinct functions encoded as power in different frequency bands. As shown, designing a symmetric network of two coupled oscillators can implement a general second-degree polynomial equation $a(P_1^2 + P_2^2) + b(P_1 + P_2) + c(P_1P_2) + d = 0$.[38] Based on the shape of generated subspaces, the coefficients should be positive with the same sign to create a circle or ellipse or negative with the same signs to create a hyperbola. However, our curiosity to utilize ORNs for realizing a broader array of functions, characterized by a richer subspace, encouraged us to explore additional topologies. These topologies bring greater complexity to the network configuration while including a wider array of points within the space.

Figure 2(c) shows two inductively coupled oscillators with $L_c = 10$ mH, where one device is connected to a 10 mH inductor and the other to a 20 mH inductor, creating an asymmetric topology based on employing nonequal inductor values for each device. As explained earlier, the behavior of two coupled oscillators is influenced by any alteration in the dynamics of each oscillator or the coupling element. Changing L_2 of the asymmetric configuration alters the second oscillator's oscillation voltage and hence its natural frequency introducing a frequency mismatch that affects the system behavior. Figure 2(d) demonstrates how an asymmetric topology affects the subspace curves created on the voltage heatmap of two coupled oscillators, revealing an asymmetric and non-circular relationship between P_1 and P_2. In this case, one of the device's inductors is fixed at 10 mH while the other one is changing to investigate the effect of different asymmetries on the implemented function. Scaling the inductor value of one of the oscillators allows the scaling of its associated power variable. The asymmetric topology can implement polynomial equation of $aP_1^2 + bP_2^2 + cP_1 + dP_2 + e(P_1P_2) + f = 0$. Nevertheless, the coefficients of the P_1 and P_2 terms are still going to be the same sign to create the generated subspace shapes. To address this issue and improve the functions to more diverse subspaces encompass more datapoint, another topology created by NDR devices as coupling impedance is modeled. Figure 2(e) shows the circuit schematic of a topology with an oscillator as a coupling impedance between two other oscillators. The I–V curve of the coupling oscillator differs from the I–V curves of the devices responsible for oscillation. The negative conductance of an NDR device $G(V_{osc}, P)$ is a nonlinear quantity of V_{osc}. However, the nonlinearity can be simplified to a linear approximation with respect to P as expressed by $G(V_{osc}, P) = \frac{P}{P_0}G(V_{osc}, P_0)$, where P_0 is a fixed value with certain current value in the I–V curve of the NDR device. Changing the power P modulates the I–V characteristics of the NDR device. Therefore, this linear dependence holds for power values that are above the smallest optical value that lets the device exhibit NDR behavior. Figure 2(d) visually demonstrates the effects of employing different NDR devices with modulated I–V curves by displaying the heatmaps of the band-passed filtered oscillation magnitude of two coupled oscillators at different current magnitudes of the coupling NDR device. While the heatmaps show symmetric behavior in the generated subspaces, the coefficients associated with P_1 and P_2 can have opposite signs, thereby including a larger set of points within the space.

After analyzing the effect of different coupling topologies on the computational subspace with two oscillators, we scale up to a 3×3 network of nearest-neighbor coupled oscillators to demonstrate feature map generation for image processing, as illustrated in Fig. 3(a). Each oscillator in a 3×3 ORN corresponds to a pixel position, and the ORN array slides across the input image with stride 1, visiting each 3×3 pixel patch sequentially, following the same spatial scanning procedure as a conventional 3×3 convolutional kernel. At each position, the nine-pixel intensities are mapped to optical power values that drive the nine oscillators. The coupled oscillatory dynamics produce time-domain voltage signals that, after fast Fourier transform (FFT) and bandpass filtering at a selected center frequency, generate a scalar output value. The collection of these scalar values across all spatial positions forms a 2D feature map. Unlike conventional convolution, the ORN applies a nonlinear transformation governed by the coupled differential equations with fixed device parameters, rather than a linear dot product with learned weights. Additionally, a single ORN topology generates multiple feature maps simultaneously, one per frequency band, enabling a single 3×3 ORN to replace the function of 32 separate learned Conv1 filters through spectral multiplexing. Hence, the ORN circuit is like a kernel filter that can slide through an image, mimicking the convolution process in CNN. All oscillators oscillate when exposed to light, impacting their neighboring signals through the coupling impedances. Our ORN architecture operates differently from the two dominant encoding paradigms in oscillatory neural networks: Phase Shift Keying (PSK) and Frequency Shift Keying (FSK). In PSK-based ONNs,[22,41–43] the input pattern is encoded as a delay to different neurons, and the oscillator phases evolve to stable phase states corresponding to memorized patterns. This architecture faces several limitations: the number of memorized and spurious patterns grows linearly with network size, propagation delays can distort input encoding, device-to-device variation prevents reliable synchronization, phase readout is sensitive to delays, interconnections scale as $\sim n^2$ limiting scalability, and input encoding requires additional DAC circuitry increasing peripheral power consumption. In FSK-based ONNs,[22,23,44] the element-wise difference between input and reference patterns is encoded as oscillator frequencies, and the presence or absence of synchronization indicates pattern match. While less

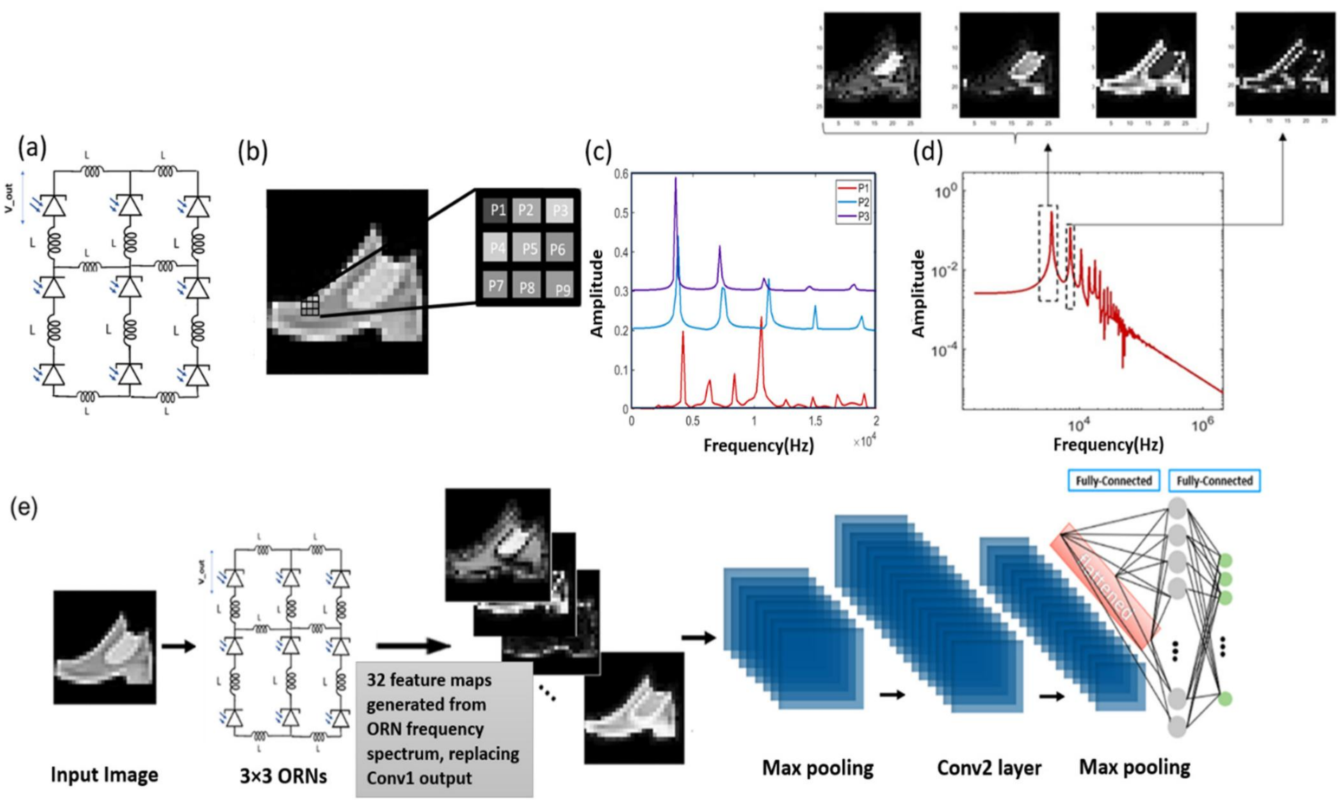

FIG. 3. Image processing with a coupled ORN network. (a) A 3*3 network of coupled oscillators inductively coupled to their nearest neighbor used as a kernel filter. (b) Image of a shoe from Fashion-MNIST dataset with the 3 × 3 ORNs swept over the whole image. (c) Frequency spectrum of the first three pixels extracted from their *V–t* signal, showing different peak frequencies related to the pixel intensities. (d) Oscillator-generated feature maps at different frequency values. (e) A CNN with its first convolutional layer replaced with the ORN-generated feature maps.

vulnerable to input distortion than PSK, FSK still requires preprocessing for frequency encoding, and device variation limits network scalability. Both architectures process data serially.

Our ORN architecture takes a different approach that addresses these limitations. The oscillators are weakly coupled, meaning they do not phase-lock or frequency-lock. Pixel intensities directly drive the oscillators through their optical power mapping without any preprocessing or encoding. The coupled oscillation dynamics generate a rich frequency spectrum where different frequency bands encode different nonlinear functions of the input in parallel. This weak coupling also makes the architecture tolerant to device-to-device variations since it does not rely on synchronization, removing one of the key scalability bottlenecks of PSK and FSK approaches.

The image shown in Fig. 3(b) is one of the gray scale images of the Fashion-MNIST (FMNIST) dataset considered an input image for the ORN filter. The reason for choosing the FMNIST dataset to perform the computations is that, unlike the traditional MNIST handwritten dataset, the FMNIST dataset is more complex and challenging and, therefore, a better benchmarking for testing the accuracy of the algorithms. The dataset includes 28×28 gray scale images of various fashion items like shoes, bags, and clothes, which have more details than simple digit dataset. The inference procedure involves sliding a 3×3 ORN convolution kernel filter across a 28×28 pixel input image with a stride of one, effectively exposing the filter to 784 distinct nine-pixel patterns, as depicted in Fig. 3(b). The V–t signals are recorded for all nine devices of the oscillatory network for each pattern. Different pixel values mean different light intensities exposed to each device resulting in different voltage amplitudes. The stored V–t signals are then transformed to frequency domain with fast Fourier transform (FFT) process. The frequency resolution of the FFT, which demonstrates how distinct the frequency components can be, is chosen based on the sampling rate and total duration of the time-domain simulated signal. Each frequency value of the transformed time-domain signal can represent a feature map of the original image. Figure 3(c) shows the FFT of three different pixel inputs, P_1, P_2, and P_3, magnified in Fig. 3(b). The pixel intensities, ranging from 0 and 1 value, act like various conductance values that multiply to the I–V curve of the NDR device to change oscillator behavior based on the light illustration on the device. As discussed earlier, Fig. 3(c) follows the decreasing trend of center and harmonic frequencies with increasing pixel intensities. Figure 3(d) shows the filtered images acquired at different frequency values of 3, 3.4, 3.6, and 7.2 kHz with 200 Hz BW. Each filtered image is a feature map containing specific information about the input image, showcasing various operations. Different feature maps demonstrate various operations like intensity filtering and image sharpening. The extracted feature at 7.2 kHz demonstrates edge detection behavior. The choice of the bandwidth is also essential for the generated feature maps and the corresponding information they provide. As we know, CNN's first convolutional layer also generates feature maps of the input image based on the specific filters available in the Conv1 layer.

Figure 3(e) shows the schematic of a CNN where the output of its first convolutional layer has been replaced with the ORN-generated feature maps. In this hybrid ORN–CNN architecture, all parameters of the ORN block including inductor values, intrinsic capacitance, NDR device characteristics, and bandpass filter parameters are fixed and not subject to backpropagation. Only the downstream CNN layers (Conv2, batch normalization, Fc1, Fc2) are trained via standard backpropagation on the Fashion-MNIST dataset. The fully software-defined CNN is designed using the PyTorch library. The model is composed of two convolutional layers each with 32 and 64, 3×3 kernel filters, respectively. The first convolutional layer takes a single-channel 28×28 gray scale image and applies 32 filters with padding 1. This is followed by batch normalization and a ReLU activation function to introduce non-linearity. A max-pooling layer with 2×2 kernel and stride of 2 follows the ReLU activation function. The max-pooling layer reduces the spatial dimensions of the first convolutional layer's feature maps by half. The second convolutional layer with 64 filters increases the depth of the feature maps and captures more detailed properties of the input feature maps. This layer is also followed by batch normalization, ReLU activation, and max pooling. Before transition from convolutional layers to fully connected layers, fc1 and fc2, the feature maps are flattened. The first fully connected layer has 64 neurons. The final fully connected layer has ten neurons, corresponding to the number of FMNIST dataset classes.

To evaluate the recognition accuracy of a combined ORN–CNN structure, we initially need to choose a specific network topology of coupled ORNs. We have systematically explored various oscillator topologies for their potential to substitute Conv1 filters, explicitly focusing on their train and test accuracies. However, given the computational resources and time required to train and test the FMNIST dataset comprising 70 000 images, we opted for a subset of 1000 images as the input dataset. With the reduction in dataset size, we ensure the simulations remain computationally feasible, allowing us to explore different oscillator topologies. Figure 4(a) presents test and train accuracy for a fully software-defined CNN trained on a reduced dataset with 800 images for training and 200 images for the test dataset. Figure 4(b) showcases the test and train accuracy of one representative ORN topology when used to replace the Conv1 layer. To further study the variability across different topologies, 30 different topologies were examined. The topologies include 3×3 asymmetric and symmetric inductively coupled oscillators, each with different sets of inductor values, 3×3 network of oscillators coupled with NDR devices, each with distinct I–V curves and inductor values, and 3×3 hybrid structures that combine NDR devices and inductors as coupling elements. Figures 4(c) and 4(d) demonstrate a histogram indicating the train and test accuracy of different topologies. As shown in Fig. 4(d), the majority of test accuracies cluster around the 71%–74% range indicating that most topologies achieve moderate performance. However, a few topologies show higher test accuracy around 77%, indicating that specific topologies perform slightly better. Hence, the variability in the test accuracy highlights that topology selection impacts the performance of the model, with some being more effective than others. The similarity in test accuracies across various topologies can be justified based on their frequency spectrum. Despite the difference among the frequency spectrum of various topologies, the extracted feature maps are approximately similar. The similarity is due to the

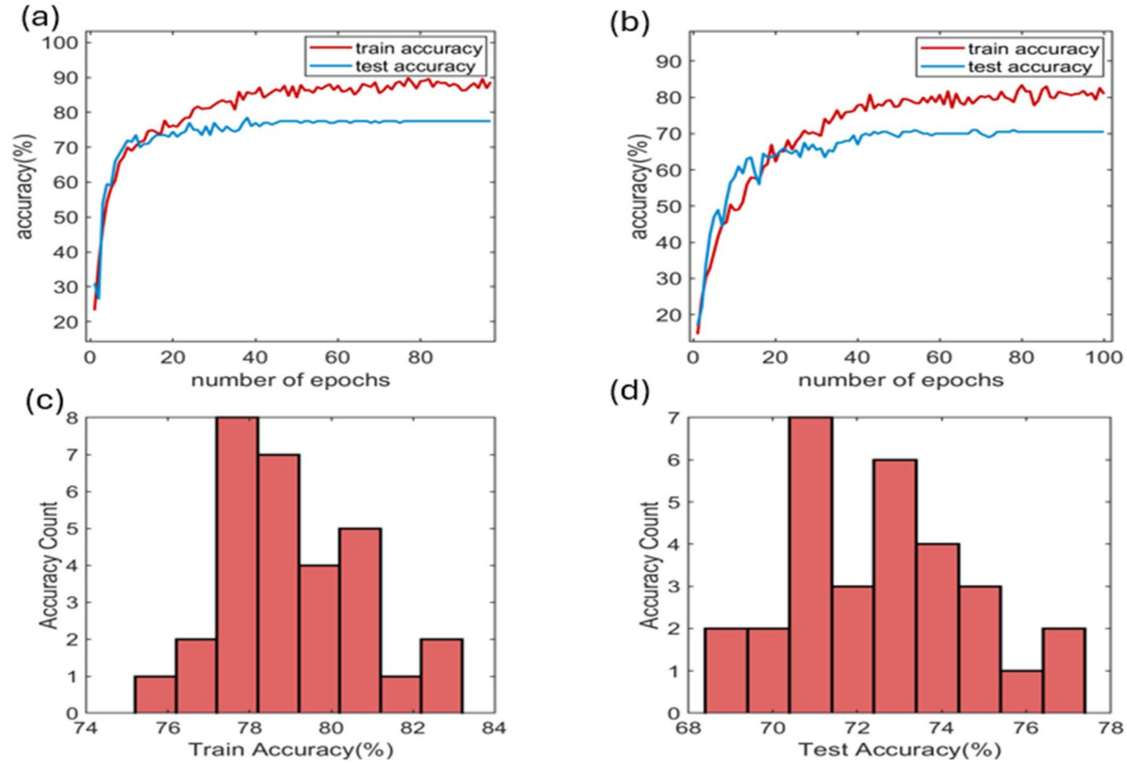

FIG. 4. Different circuit topologies analysis. (a) Test and train accuracy of a fully software-defined CNN for 1000 FMNIST images. (b) Test and train accuracy of one representative ORN topology when used to replace the Conv1 layer. (c) and (d) Histogram of Conv1 replaced model, indicating train and test accuracies across 30 different topologies.

interconnected nature of subpeaks that are not significantly discrete, leading to similar feature maps and, hence, similar accuracy results across different topologies. Interestingly, the histogram suggests that over half of the analyzed topologies, showing 71%–74% accuracies, are viable for replacing the Conv1 layer. However, the varying accuracies compel us to choose a topology that consistently demonstrates high accuracy with little accuracy drop in its architecture. Among the evaluated topologies, the asymmetric inductive coupling architecture stood out. Across various inductor values, this topology consistently achieved accuracy values of 74% or higher. This consistency highlights the asymmetric inductive coupling topology as the potential candidate for replacing the Conv1 layer. For further analysis, 70 000 images of the FMNIST dataset have been processed using both a conventional CNN and a 3×3 asymmetric inductively coupled ORN, with 60 000 images for the training dataset and 10 000 for the testing dataset. The conventional CNN analyses the data and calculates the train and test accuracy for 50 epochs. In addition, the ORN–CNN shown in Fig. 3(e) replaces the output of Conv1 for all 70 000 images of the FMNIST dataset and calculates train and test accuracy.

Figures 5(a) and 5(b) demonstrate a comparison between both CNN and ORN–CNN test and train accuracy, revealing similar test accuracy levels of 93% for the fully software-defined case and around 92% for the CNN structure with ORNS block as its first convolutional layer. The successful integration of an ORN network as a replacement for the first layer of a CNN demonstrates that an ORN network is capable of capturing essential features to make accurate predictions of a large and complex input dataset with a 3×3 filter topology rather than using 32 different filters of a traditional CNN. This finding challenges the prevailing assumptions that only convolutional layers are effective in feature analysis in deep learning models. Therefore, our oscillatory network appears to be an alternative structure that achieves comparable results with significantly low-power consumption.

As mentioned previously, according to the voltage range shown in the NDR device I–V curve in Fig. 1(b), the device does not require any external electrical power for oscillation. However, since ultimately, we want to implement our software-based calculations into a hardware platform, we might need some peripheral circuitry for converting the voltage–time oscillations into a frequency spectrum through amplifiers, a sample and hold circuit, and a DFT processor. As shown earlier, different frequency bands of the FFT represent a distinctive feature map. The detailed calculation for energy consumption of an ORN considering an eight-bit integer FFT processor is discussed in Ref. 38. A 3×3 ORN

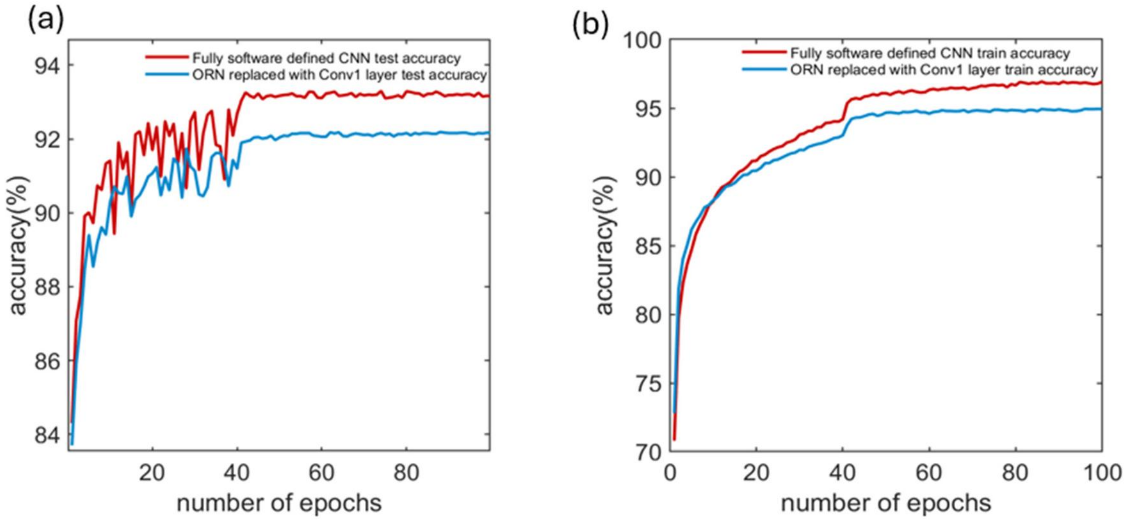

FIG. 5. Test and train accuracy comparison. (a) Test accuracy of a fully software-defined CNN, and a CNN with its first convolutional layer substituted by an ORN for 70 000 FMNIST images. (b) Train accuracy for a fully software-defined CNN, and a CNN with its first convolutional layer substituted by an ORN for 70 000 FMNIST images.

performs image processing with 42 211 TOPS/W, equivalent to the energy consumption of 24 aJ/OP.[38] Discussing the energy consumption of the integrated CNN with ORNs highlights the potential benefits of replacing convolutional layers of a CNN with a network of coupled oscillators characterized by their significantly low-power consumption. The energy consumption of the implemented CNN with two convolutional layers, each with 32 and 64 filters, two pooling layers, and three fully connected layers, is as follows. For convolutional layers, the number of operations can be estimated as follows:

$$\text{Operation per layer} = K^2.C_{\text{in}}.H_{\text{out}}.W_{\text{out}}.C_{\text{out}}, \qquad (14)$$

where K is the kernel size, C_{in} and C_{out} are the numbers of input and output channels, and H_{out} and W_{out} are the dimensions of the output feature maps. The first convolutional layer of the CNN operates multiply and accumulation operations (MACs) on an input image of size 28×28 with a 3×3 kernel, one input channel, and 32 output channels and creates 28×28 feature maps. Hence, the number of operations for the first convolutional layer is 225 792. The second convolutional layer with 32 input channels convolving feature maps of size 14×14 due to a pooling layer and 64 output channels lead to 3 612 672 operations. In addition, the CNN has two fully connected layers whose computational load is derived from the product of the number of input and output channels, resulting in 200 704 and 640 operations for Fc1 and Fc2, respectively. The overall MAC operations of the designed CNN amount to 4 039 778 translating to 8 079 556 floating point operations (FLOPs). Utilizing the GPU NVIDIA RTX A5000 that runs 222.2 TFLOPS/s at 230 W, the total energy consumption accounts for 8.32 μJ with 2.05 pJ/op. Given this, the energy consumption carried out by the first convolutional layer is 0.46 μJ. In our ORN circuitry, the FFT processor generates 64 frequency channels and each convolution operation for a 3×3 kernel is equivalent to 17 operations leading to an overall 1088 MAC operations. Therefore, the total energy consumption of the ORNs' peripheral circuit is 26.112 fJ. This calculation underscores the possible advantages of integrating an energy-efficient network of coupled oscillators with a CNN specially for applications where power consumption is a critical constraint.

To highlight the impact of employing a hybrid ORN-software CNN structure on power consumption, we evaluate different scenarios to investigate potential power reductions. Figure 6 shows the energy consumption comparison across the four scenarios. In scenario 1, a 3×3 photosensitive neuron block is replaced with 32 3×3 kernel filters in the first convolutional layer, capable of generating diverse feature maps with an energy consumption of 20.47 pJ. This layer is followed by an ADC block with a 10 fJ/conversion energy cost. Considering an eight-bit ADC, the energy consumption per number is equivalent to 2.56 pJ. With the ADC block processing 32 maxpooled feature maps, the resulting energy consumption is 16.056 nJ. The subsequent layers, Conv2, Fc1, and Fc2 consume 7.44, 0.413, and 1.3 nJ, respectively. The overall energy consumption for this scenario amounts to 7.853 μJ. In second scenario, both Conv1 and Conv2 layers are replaced by ORNs. The ORNs energy consumption for the first and second convolutional layers are 20.47 and 163.76 pJ, respectively. The ADC processes $64 \times 7 \times 7 = 3136$ inputs, resulting in an energy consumption of 8.028 nJ. The first and second fully connected layers consume 0.413 μJ and 1.3 nJ, respectively. The total energy consumption for this scenario is 422.32 nJ. In scenario 3, both Conv1 and Conv2 are replaced

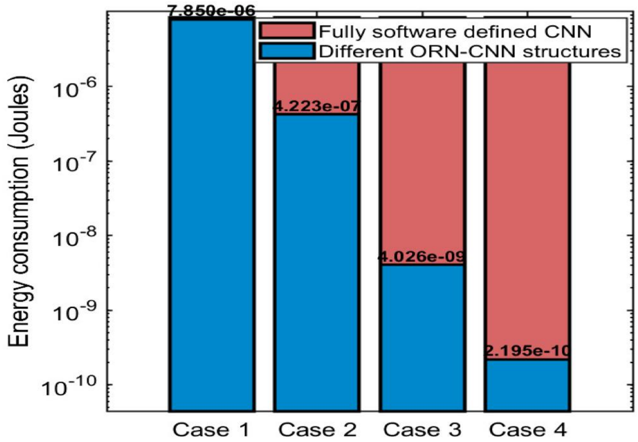

FIG. 6. Energy cost of different hybrid ORN–CNN structures, case 1: ORNs is replaced with the Conv1 layer of a CNN, case 2: both Conv1 and Conv2 layers are replaced by ORNs, case 3: both Conv1 and Conv2 are replaced with ORNs, followed by a memristor layer, and case 4: Conv1, Conv2, and both fully connected layers are substituted by ORNs.

with ORNs, followed by a memristor layer. The memristor, implemented with an eight-bit SONOS memory operating at ten TOPS/W, consumes 10 fJ/op, resulting in an energy consumption of 4.026 nJ. Therefore, the energy consumption of a memristor is equal to 4.026 nJ. The next layer, an eight-bit ADC, consumes an additional 25.6 pJ. Therefore, the total energy consumption of the third scenario is equivalent to 4.026 nJ. In the fourth scenario, Conv1, Conv2, and both fully connected layers are substituted by ORNs. The ORNs consume 20.47 and 164.76 pJ for the first and second convolutional layers, respectively. Using ORNs to replace fully connected layers results in an energy consumption of 9.66 pJ. The ADC layer in this configuration consumes 25.6 pJ. Consequently, the total energy consumption for this scenario is 219.49 pJ.

The detailed power consumption calculations for different cases presented above emphasize the transformative potential of integration of ORNs blocks with software-defined layers for significant power reduction. These calculations validate the feasibility of ORNs in real-world applications and show a quantitative basis for comparing traditional architectures with hybrid approaches. The significant contrast in energy consumption between scenarios demonstrates the critical role of optimized design choices in achieving ultra-low-power operation, particularly in energy-constrained systems.

As an additional benchmark, we compare our results with the state-of-the-art NVIDIA H100 SXM GPU, which achieves 3958 TOPS at INT8 precision with a TDP of 700 W, corresponding to 177 fJ per INT8 operation. The total CNN energy on the H100 amounts to 0.715 μJ, with the first convolutional layer consuming approximately 40 nJ. Our ORN replacement consumes 26.1 fJ for the equivalent Conv1 computation, over six orders of magnitude lower. This advantage arises from both lower energy per operation (24 aJ vs 177 fJ) and fewer total operations (1088 vs 225 792) since a single ORN topology generates multiple feature maps simultaneously through spectral multiplexing. We note that the GPU figure includes full chip-level overhead while the ORN figure reflects projected peripheral circuit energy only.

CONCLUSION

In conclusion, our study demonstrates the potential of implementing an ultra-low power network of coupled oscillators as a substitution of the convolutional layer filters of a CNN. The results reveal that the integration of ORNs with a CNN maintains the recognition accuracy levels of an original CNN architecture. Moreover, this network of oscillators can perform computational tasks with remarkable energy efficiency while consuming a mere 24 aJ per operation, over three orders of magnitude more efficient per operation than the state-of-the-art NVIDIA H100 GPU (177 fJ per INT8 operation). The energy projection of different scenarios for replacing CNN layers further highlights the benefit of replacing Conv1, Conv2, and both fully connected layers lead to total energy consumption as low as 219 pJ for a CNN. Hence, showing that ORNS can be a power-saving candidate to replace traditional convolutional layers in CNNS, particularly in applications where the main priority is energy conservation. The dependence of computational subspace on coupling impedance values suggests an intriguing pathway toward hardware-level trainability. If the coupling and device inductors were implemented as tunable elements, the network's nonlinear transfer function could be adjusted. Although the present study uses millihenry-range inductors in simulation, scalable chip-level realizations can employ active inductors such as the Hara inductor (a single MOSFET and a resistor). Additionally, scaling the NDR device area reduces the intrinsic capacitance, shifting the required inductance to the nH to μH range achievable with on-chip spiral inductors in standard CMOS processes. Alternative coupling mechanisms such as capacitive or resistive networks may also be explored. Future work will focus on experimental hardware implementation of the ORN array and its readout circuitry to validate low-power inference under practical nonidealities such as noise and device-to-device variations.

SUPPLEMENTARY MATERIAL

See the supplementary material for additional details regarding the methodology for replacing the oscillator-generated feature maps with the first convolutional layer feature maps using SSIM analysis.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Seyedeh Atiyeh Abbasi Jalal: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Ragib Ahsan:** Conceptualization (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **Zezhi Wu:** Investigation (equal); Validation (equal); Writing – review & editing (equal). **Mirbehrad Mousavi:** Investigation (equal); Validation (equal); Writing – review & editing (equal). **Rehan Kapadia:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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